

Statistical Impossibility and Possibility of Aligning LLMs with Human Preferences

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Collaborators and References

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Background: Training Workflow of Large Language Models (LLMs)

Pretraining The model is trained by predicting the next token on large-scale unlabeled text through self-supervised learning.

This stage gives the model broad linguistic and factual knowledge.

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Alignment The model is aligned with human values using preference data.

This improves the helpfulness and safety of responses from the model.

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Human Preferences Oracle

$\mathcal{P}(y \succ y')$ is defined as the expected fraction of individuals who prefer y over y' . It follows that $\mathcal{P}(y \succ y') + \mathcal{P}(y' \succ y) = 1$, and we define $\mathcal{P}(y \succ y) = 1/2$. For any distinct responses y and y' , we say that y beats y' if $\mathcal{P}(y \succ y') > 1/2$.

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- **No-tie:** For any distinct responses y and y' , we assume that $\mathcal{P}(y \succ y') \neq 1/2$.

Reinforcement Learning from Human Feedback (Ouyang et al. 2022)

- Human preferences are typically modeled using the BT model, which assumes the existence of a reward model $r(x, y)$ such that:

$$\mathcal{P}(y \succ y' \mid x) = \frac{\exp(r(x, y))}{\exp(r(x, y)) + \exp(r(x, y'))}.$$

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- In practice, through collecting an offline dataset of comparisons D_{comp} , we can learn a reward model via maximum likelihood:

$$\max_r \mathbb{E}_{(x, y_w, y_l) \sim D_{\text{comp}}} [\log(\sigma(r(x, y_w) - r(x, y_l)))],$$

where $\sigma(u) := 1/(1 + \exp(-u))$ is the logistic function.

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- The standard RLHF maximizes the reward to align LLMs with human preferences.

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What is the limit of modeling human preference using reward?

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Consistency

A reward model $r : \mathcal{Y} \rightarrow \mathbb{R}$ is consistent with a preference $\mathcal{P}(y \succ y')$ if for any two distinct responses y and y' , $r(y) > r(y')$ implies $\mathcal{P}(y \succ y') > 1/2$.

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- This definition encompasses general reward models beyond the BT model. For example, any fixed positive and nondecreasing function Ψ induces the following reward model

$$\mathcal{P}(y \succ y' \mid x) = \frac{\Psi(r(x, y))}{\Psi(r(x, y)) + \Psi(r(x, y'))}.$$

Reward Models can Only Capture Transitive Preferences

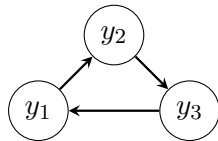


Figure 1: Condorcet paradox (Gehrlein 2006): y_1 beats y_2 , y_2 beats y_3 , y_3 beats y_1 .

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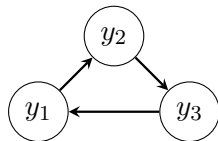


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Condorcet Cycle

A sequence of responses $y_{i_1}, \dots, y_{i_r}$ forms a Condorcet r -cycle if $\mathcal{P}(y_{i_p} \succ y_{i_{p+1}}) > 1/2$ for all $p = 1, \dots, r$, where $y_{i_{r+1}} := y_{i_1}$ by convention.

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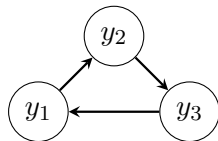


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Theorem (Reward-Consistent Preferences $\iff$ Acyclic Preferences)

For any set of responses $\{y_1, \dots, y_n\}$ with $n \geq 3$ and any preference $\mathcal{P}(y \succ y')$ defined on this set, there exists a reward model $r(y)$ that is consistent with the preference $\mathcal{P}(y \succ y')$ if and only if there is no Condorcet cycle in the set of responses.

Linear Preference Ranking

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Statistical Assumptions on Human Preferences

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Luce Model

Each individual independently samples a linear preference ranking from $\text{Luce}(\lambda_1, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n$ are some positive real weights. Mathematically, for any permutation $\{i_1, \dots, i_n\}$ of $\{1, \dots, n\}$, the probability of each linear preference is given by:

$$\mathbb{P}(y_{i_1} \succ \dots \succ y_{i_n}) = \frac{\lambda_{i_1}}{\lambda_{i_1} + \dots + \lambda_{i_n}} \times \frac{\lambda_{i_2}}{\lambda_{i_2} + \dots + \lambda_{i_n}} \times \dots \times \frac{\lambda_{i_{n-1}}}{\lambda_{i_n} + \lambda_{i_{n-1}}}.$$

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- When $\lambda_1 = \dots = \lambda_n$, the Luce model reduces to the popular **impartial culture (IC) condition** in social choice theory (Guilbaud 1952; May 1971; Bell 1978; Gehrlein 2006).

Probability of Condorcet Cycles

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Theorem ($\mathbb{P}(\text{Condorcet Cycles}) \rightarrow 1$ Exponentially Fast)

Let the number of responses $n \geq 3$ and the number of labelers $m \geq 3$. In addition, suppose the weights are bounded with $\lambda_i \in [A, B]$, where $B \geq A > 0$. Then,

$$\mathbb{P}_{m,n}(\text{Condorcet cycle}) \geq 1 - c_1 e^{-c_2 n},$$

where $c_1, c_2 > 0$ are constants that only depends on m, A, B .

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- Under the IC condition, c_1 and c_2 can be chosen to be independent of m .

Lemma (Equivalent Formulation of the Luce Model)

The Luce model $\text{Luce}(\lambda_1, \dots, \lambda_n)$ is equivalent to the preference generation scheme:

- 1 Sample n scores from independent exponential random variables U_k with parameters λ_k , denoted by $\{U_k \sim \text{Exp}(\lambda_k)\}_{1 \leq k \leq n}$;
- 2 Order the scores from small to large as $U_{i_1} < \dots < U_{i_n}$; the preference is then given by $y_{i_1} \succ \dots \succ y_{i_n}$.

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- For each group of three, the probability of the occurrence of a Condorcet triangle is lower bounded by some constant that only depends on m, A, B .
- Under IC, the constant can be made independent of m using the CLT.

Which Response Does RLHF Outputs?

Theorem (Siththaranjan, Laidlaw, and Hadfield-Menell 2024)

- Assume that the reward is learnt by maximizing likelihood under perfect data, then RLHF implements the famous *Borda count* in social choice theory.

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- RLHF always collapse to the single response, known as *preference collapse* (Xiao, Li, et al. 2025).
- The Borda count does not satisfies common social choice theory axioms.

Nash Learning from Human Feedback (NLHF)

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- **Non-reward-based:** NLHF is a non-reward-based LLM alignment approach that directly leverages a preference model. In practice, the preference model is trained using a preference dataset D_{comp} .

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$$\pi^* := \arg \max_{\pi} \min_{\pi'} \mathbb{E}_{x \sim \rho} \left[\mathbb{E}_{y \sim \pi(\cdot|x), y' \sim \pi'(\cdot|x)} \mathcal{P}(y \succ y' | x) \right]. \quad (1)$$

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Can NLHF avoid preference collapse?

Table 1: Payoff matrix with three responses $\{y_1, y_2, y_3\}$.

$\mathcal{P}(y \succ y')$	$y' = y_1$	$y' = y_2$	$y' = y_3$
$y = y_1$	0.5	α_{12}	$1 - \alpha_{31}$
$y = y_2$	$1 - \alpha_{12}$	0.5	α_{23}
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- Assume $\alpha_{12} > 0.5$, $\alpha_{23} > 0.5$, $\alpha_{31} > 0.5$.
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- Assume $\alpha_{12} > 0.5$, $\alpha_{23} > 0.5$, $\alpha_{31} > 0.5$.
- This preference structure contains a Condorcet cycle $y_1 \succ y_2 \succ y_3 \succ y_1$.
- In this case, no pure strategy Nash equilibrium exists, and the Nash equilibrium must be mixed, with positive probability assigned to all three responses.

Observation II

Table 2: Payoff matrix with four responses $\{y_1, y_2, y_3, y_4\}$.

$\mathcal{P}(y \succ y')$	$y' = y_1$	$y' = y_2$	$y' = y_3$	$y' = y_4$
$y = y_1$	0.5	0.51	0.46	0.47
$y = y_2$	0.49	0.5	0.51	0.48
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- This preference structure contains a Condorcet cycle $y_1 \succ y_2 \succ y_3 \succ y_1$.
- However, the pure strategy (y_4, y_4) is a Nash equilibrium in pure strategies.

Condorcet Winning Response

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- Maura-Rivero et al. (2025) relates the second result to *maximal lottery* (Fishburn 1984) in social choice theory.

Probability of Condorcet Winning Response

Theorem ($\mathbb{P}(\text{Condorcet Winning Response}) \rightarrow 0$)

Let the number of responses be n and the number of labelers $m \geq 3$, and let $l = \lceil \frac{m}{2} \rceil$. In addition, suppose the weights are bounded with $\lambda_i \in [A, B]$, where $B \geq A > 0$. Then

$$\mathbb{P}_{m,n}(\text{Condorcet winning response}) = \tilde{\Theta} \left(n^{1-\frac{m}{l}} \right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$\tilde{\Theta}$ hides logarithm factors in n , and constant that only depends on m, A, B .

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- Sauermann (2022) obtained the same rate under the restrictive IC condition.

Probability of Condorcet Winning Response

Theorem ($\mathbb{P}(\text{Condorcet Winning Response}) \rightarrow 0$)

Let the number of responses be n and the number of labelers $m \geq 3$, and let $l = \lceil \frac{m}{2} \rceil$. In addition, suppose the weights are bounded with $\lambda_i \in [A, B]$, where $B \geq A > 0$. Then

$$\mathbb{P}_{m,n}(\text{Condorcet winning response}) = \tilde{\Theta} \left(n^{1-\frac{m}{l}} \right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$\tilde{\Theta}$ hides logarithm factors in n , and constant that only depends on m, A, B .

- Sauermann (2022) obtained the same rate under the restrictive IC condition.
- Her proof is algebraic in nature and cannot be extended to general Luce model.

Proof Idea

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$$N_j = \{i \in \{2, \dots, n+1\} \mid U_i^j < U_1^j\}.$$

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- We further define the set $T_t := N_{j_1} \cap \dots \cap N_{j_l}$.
- y_1 is the Condorcet winning response $\iff$ there does not exist any $i \neq 1$ such that $U_i^j < U_1^j$ for more than $l = \lceil \frac{m}{2} \rceil$ labelers:

$$\mathbb{P}(y_1 \text{ is the Condorcet winning response}) = \mathbb{P}\left(\bigcap_{t \in \mathcal{T}} \{T_t = \emptyset\}\right).$$

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- Conditioning on $\{X_j\}$, N_j are independent uniformly random set with $|N_j| = X_j$.
- This scheme can be approximated by sampling each i independently with probability X_j/n .
- Thus $|N_{(1)} \cap \dots \cap N_{(l)}| \approx \text{Binomial}\left(n, \frac{X_{(1)} \dots X_{(l)}}{n^l}\right)$

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- $X_1, \dots, X_m$ are i.i.d. uniformly distributed on $\{0, \dots, n\}$, and

$$\mathbb{P}(X_{(1)} \cdots X_{(l)} \leq n^{l-1}(\log n)^2) \leq \mathbb{P}(X_1 \cdots X_m \leq \tilde{O}(n^{(l-1)\frac{m}{l}})) \lesssim \tilde{O}(n^{-\frac{m}{l}}).$$

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- Similar intuitions hold in the Luce model, although there is much more technicality.

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- The idea is to use the inclusion-exclusion principle

$$\mathbb{P} (Z_2 = 0, \dots, Z_{n+1} = 0) = 1 - \sum_{k=1}^n (-1)^{k-1} \sum_{\{i_1, \dots, i_k\} \subseteq \{2, \dots, n+1\}} \mathbb{P} (Z_{i_1} = 1, \dots, Z_{i_k} = 1)$$

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- Using the union bound,

$$\mathbb{P} (Z_2 = 1, \dots, Z_{k+1} = 1) \leq \sum_{t_1, \dots, t_k \in \mathcal{T}} \mathbb{P} (2 \in T_{t_1}, \dots, k+1 \in T_{t_k}) .$$

Lower Bound II

- Consider the following critical profile

$$\mathbf{B} := \left\{ \mathbf{U}_1 \in \mathbb{R}^m : 0 \leq U_1^j \leq \frac{1}{B} \cdot \frac{1}{(2n)^{\frac{1}{t}} \binom{m}{l}^{\frac{1}{t}}}, \forall j = 1, \dots, m \right\}.$$

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- This profile has exactly the rate we seek

$$\mathbb{P}(\mathbf{U}_1 \in \mathbf{B}) = \prod_{j=1}^m \left(1 - \exp \left(- \frac{\lambda_1}{B \binom{m}{l}^{\frac{1}{l}} (2n)^{\frac{1}{l}}} \right) \right) \gtrsim \frac{1}{n^{\frac{m}{l}}}.$$

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- The conditional probability can be easily bounded when $\mathbf{U}_1 \in \mathbf{B}$:

$$\mathbb{P}(2 \in T_{t_1}, \dots, k+1 \in T_{t_k} \mid \mathbf{U}_1) = \prod_{s=1}^k \prod_{j \in t_s} \left(1 - e^{-\lambda_{s+1} U_1^j} \right) \leq \frac{1}{\binom{m}{l}^k} \left(\frac{1}{2n} \right)^k.$$

Lower Bound III

- Finally

$$\mathbb{P}(Z_2 = 0, \dots, Z_{n+1} = 0) \geq \mathbb{P}(Z_2 = 0, \dots, Z_{n+1} = 0 \mid \mathbf{U}_1 \in \mathbf{B}) \mathbb{P}(\mathbf{U}_1 \in \mathbf{B}) .$$

Lower Bound III

- Finally

$$\mathbb{P}(Z_2 = 0, \dots, Z_{n+1} = 0) \geq \mathbb{P}(Z_2 = 0, \dots, Z_{n+1} = 0 \mid \mathbf{U}_1 \in \mathbf{B}) \mathbb{P}(\mathbf{U}_1 \in \mathbf{B}) .$$

- The proof is completed by combining the inclusion-exclusion principle and the following lemma:

Technical Lemma

There exists a universal constant $0 < c < 1$ such that for all integers $n \geq 1$,

$$1 - \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n}{2k+1} \left(\frac{1}{2n}\right)^{2k+1} \geq c .$$

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Theorem ($\mathbb{P}(\text{Condorcet Winning Response}) \rightarrow 0$)

Let the number of responses be n and the number of labelers be $m \geq 3$, and define $l = \lceil m/2 \rceil$. For $0 \leq k < m - l$, suppose that $m - k$ individuals follows the Luce model with bounded weights $\lambda_i \in [A, B]$, where $B \geq A > 0$, and the remaining k individuals have arbitrary linear preference ranking independent of the first $m - k$ individuals. Then

$$\mathbb{P}_{m,n}(\text{Condorcet winning response}) \rightarrow 0.$$

A Summary

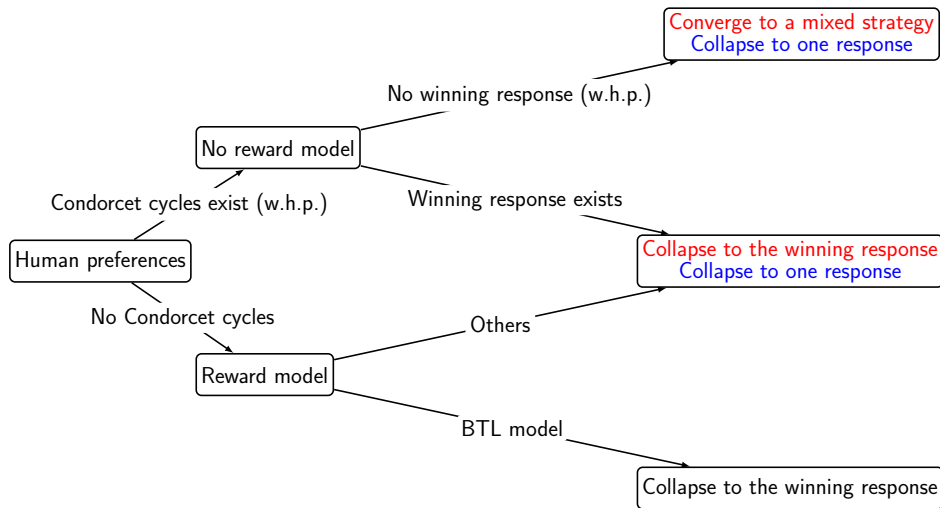


Figure 2: Red text indicates properties of NLHF, while blue text indicates properties of RLHF.

Other Social Choice Theory Axioms: Smith Consistency and Beyond

Which policy does NLHF employ if there is no winning response?

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Theorem (Hierarchy of Responses)

The set of responses can be partitioned into disjoint subsets $S_1, \dots, S_k$ such that:

- 1 Each S_i either forms a Condorcet cycle or is a single response.
- 2 For any $j > i$, any response $y \in S_i$ and $y' \in S_j$, $\mathcal{P}(y \succ y') > \frac{1}{2}$.

Moreover, this decomposition is unique. S_1 is referred to as the Smith Set.

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Theorem (Smith Consistency)

NLHF is **Smith consistent** (Shoham and Leyton-Brown 2008), in the sense that the support of any Nash equilibrium is contained in the Smith set S_1 .

General Game-Theoretic Alignment

- In general, we can consider the following objective:

$$\pi^* := \arg \max_{\pi} \min_{\pi'} \mathbb{E}_{x \sim \rho} \left[\mathbb{E}_{y \sim \pi(\cdot|x), y' \sim \pi'(\cdot|x)} \left[\Psi(\mathcal{P}(y \succ y' \mid x)) \right] \right] . \quad (2)$$

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Table 3: The necessary and sufficient condition on continuous Ψ to guarantee certain desirable alignment axioms.

Condorcet consistency	$\Psi(t) \geq \Psi(1/2), \forall 1/2 \leq t \leq 1$ and $\Psi(t) < \Psi(1/2), \forall 0 \leq t < 1/2$
- Condorcet consistency & Mixed	$\Psi(t) + \Psi(1-t) \geq 2\Psi(1/2), \forall 1/2 \leq t \leq 1$ and $\Psi(t) < \Psi(1/2), \forall 0 \leq t < 1/2$
Smith consistency	$\Psi(t) + \Psi(1-t) = 2\Psi(1/2), \forall 1/2 \leq t \leq 1$ and $\Psi(t) < \Psi(1/2), \forall 0 \leq t < 1/2$
Preference Matching	No Ψ exists

Preference Matching Policy

A policy is said to be a preference matching policy, if for every prompt x and any pair of responses y, y' ,

$$\frac{\pi(y \mid x)}{\pi(y' \mid x)} = \frac{\mathcal{P}(y \succ y' \mid x)}{\mathcal{P}(y' \succ y \mid x)}.$$

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$$\pi^*(y \mid x) = \frac{\exp(r(x, y))}{\sum_{y'} \exp(r(x, y'))}.$$

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- Xiao, Li, et al. (2025) showed that RLHF can achieve preference matching by tailoring a regularization term.

Additional Topics: What Happens when Regularization is Added?

NLHF with Regularization (Munos et al. 2024)

In practice, an additional KL regularization term is often added into the NLHF objective:

$$\max_{\pi} \min_{\pi'} \mathcal{P}(\pi \succ \pi') - \tau \left(\mathbb{E}_{x \sim \rho} [\text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) - \text{KL}(\pi'(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))] \right),$$

where $\tau > 0$ is a parameter controlling the deviation from the base reference policy π_{ref} .

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Theorem (An Equivalent Objective)

The above problem is equivalent to:

$$\begin{aligned} \max_{\pi} \mathbb{E}_{x \sim \rho} [\mathcal{Z}(x, \pi) - \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))], \\ \mathcal{Z}(x, \pi) := -\log \mathbb{E}_{y' \sim \pi_{\text{ref}}(\cdot | x)} \left[\exp \left(-\frac{1}{\tau} \cdot \mathbb{E}_{y \sim \pi(\cdot | x)} [\mathcal{P}(y \succ y' | x)] \right) \right]. \end{aligned} \quad (3)$$

Algorithm: Nash-MD (Munos et al. 2024)

Nash-MD

Nash-MD solves the regularized NLHF using the following double-loop algorithm:

$$\pi_{t+1}(\cdot | x) = \arg \max_{\pi(\cdot | x)} \left\{ \eta_t \cdot \mathcal{P} \left(\pi \succ \pi_t^{\text{ref}} \right) - \text{KL} \left(\pi(\cdot | x) \| \pi_t^{\text{ref}}(\cdot | x) \right) \right\} ,$$

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- Nash-MD produces a sequence of policies $\{\pi_t\}_{1 \leq t \leq T}$ with last-iterate convergence (in KL-divergence) to the regularized Nash equilibrium at a speed $O(1/T)$.
- In practical deep learning architectures, **Nash-MD** employs a general form of policy gradient for model training. This approach relies on the double-loop formulation of Nash-MD, but it does not achieve exact maximization at each iteration, potentially introducing an unknown bias into the optimization process.

General Form of Policy Gradient in Nash-MD

$$\mathbb{E}_{y \sim \pi_\phi(\cdot|x), y' \sim \pi'(\cdot|x)} \left[\nabla_\phi \log \pi_\phi(y | x) \cdot \left(\mathcal{P}(y \succ y' | x) - \tau \log \frac{\pi_\phi(y | x)}{\pi_{\text{ref}}(y | x)} \right) \right]$$

- The policy π' is:
 - 1 the geometric mixture policy $\pi'(y | x) \propto \pi_\phi(y | x)^{1-\beta} \pi_{\text{ref}}(y | x)^\beta$ in Nash-MD.
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- 1 the geometric mixture policy $\pi'(y | x) \propto \pi_\phi(y | x)^{1-\beta} \pi_{\text{ref}}(y | x)^\beta$ in Nash-MD.
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Theorem (Policy Gradient of Objective (3))

$$\mathbb{E}_{y \sim \pi_\phi(\cdot|x), y' \sim \pi'_\star(\cdot|x)} \left[\nabla_\phi \log \pi_\phi(y | x) \cdot \left(\mathcal{P}(y \succ y' | x) - \tau \log \frac{\pi_\phi(y | x)}{\pi_{\text{ref}}(y | x)} \right), \right] \quad (4)$$

$$\pi'_\star(y | x) \propto \pi_{\text{ref}}(y | x) \cdot \exp \left(-\frac{\mathcal{P}(\pi_\phi \succ y | x)}{\tau} \right).$$

A Single-Loop Algorithm: Nash-RS

- We can use PPO to solve the equivalent problem (3), using the policy gradient (4).
- The only remaining issue is how to sample from $\pi'_\star$. Since it can be viewed as a Gibbs distribution, we can use rejection sampling to efficiently draw samples from $\pi'_\star$.

Algorithm (Nash-RS)

- 1 Use rejection sampling to draw samples y' from $\pi'_\star$.
- 2 Draw samples from π_ϕ and compute the policy gradient.
- 3 Use PPO for the optimization with the computed policy gradient.

Copeland RLHF

When each comparison is labeled by more than one labeler, each comparison should not be treated separately:

$$\max_r \sum_{y_i, y_j, i \neq j} \mathbb{1}[\mathcal{P}(y_i \succ y_j) > 1/2] \cdot \log(\sigma(r(y_i) - r(y_j))).$$

Additional Topics: Why does RLHF Works in Practice?

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- In reality, for any two distinct responses y, y' , exactly one human labeler provides a comparison.
- In this case, RLHF and Copeland RLHF are equivalent.

Theorem (Copeland RLHF is Condorcet Consistent)

- Assume that the reward is learnt by maximizing the above likelihood under perfect data, then RLHF implements the famous *Copeland rule* in social choice theory.

Theorem (Copeland RLHF is Condorcet Consistent)

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- Specifically, let

$$w_i = \sum_{k \neq i} \mathbb{1}[\mathcal{P}(y_i \succ y_j) > 1/2],$$

and

$$r(y_i) > r(y_j) \iff w_i > w_j.$$

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